Automated EEG Screening with a Transformer Model in Shadow Operation in Kenya

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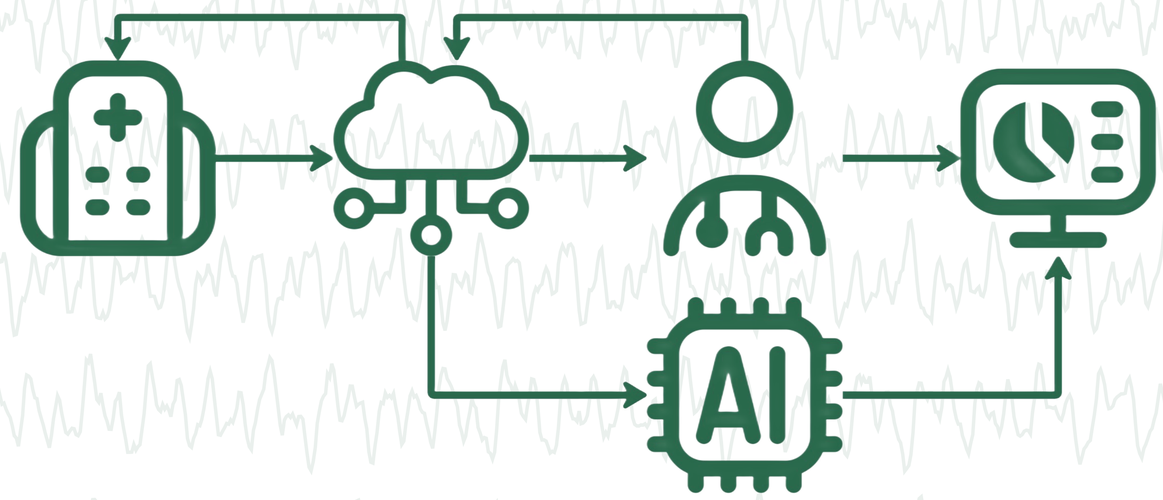
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Abstract

In many emerging markets, including Kenya, Indonesia, the Philippines, Malaysia, and Vietnam, the shortage of trained neurophysiologists means that a large number of people live with undiagnosed epilepsy. This is known as the *Epilepsy Treatment Gap*. While portable EEG solutions, like the BrainCapture (BC-1) system, have increased data collection capacity, the bottleneck remains the expert interpretation of these recordings. Artificial Intelligence (AI), particularly the use of self-supervised learning (SSL) and transformer-based models offers a scalable solution to assist technicians and clinicians.

This study evaluates the real-world performance of a transformer-based EEG screening model operating in a "shadow" capacity within a clinical workflow in Kenya, aiming to provide automated triaging and decision support in resource-limited settings.





Methodology

Deployment: An EEG transformer was integrated into the BrainCapture infrastructure across over 20 Kenyan clinical sites. During this "shadow" phase, AI results were generated in the background without influencing clinical decision-making.

Architecture & Training: Levering an encoder pretrained using an attention-based framework, inspired by [1], a classifier was fine-tuned on 2,130 recordings collected from the BrainCapture operation during the period November 2023 to October 2025.

Preprocessing & Task: Data was processed per [2] for binary triage (*abnormal* vs *normal*) using 60s windows. Abnormalities (spikes, sharp waves, slowing) were defined via the SCORE framework [3].

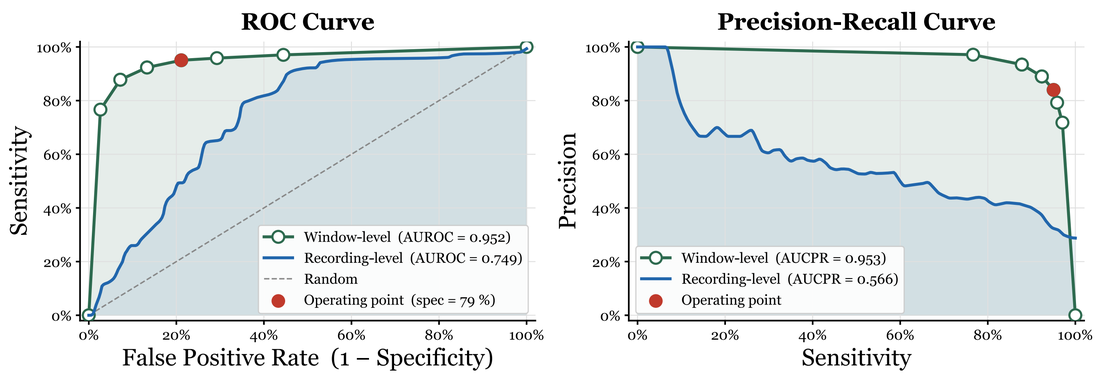
Aggregation: The 60-second window-level predictions were aggregated to the recording level using a CNN-based approach. For both window and recording level predictions, we report specificity at a clinical threshold of 95% sensitivity.

Evaluation: To evaluate the model's real-world utility, we compared model outputs to the established clinical workflow: Neurophysiology assistant annotation validated by senior neurophysiologists.

We validated the model on an evaluation dataset of 347 recordings collected from the BrainCapture operation during the period November 2025 to January 2026.

Results

With a clinical threshold of 95% sensitivity, we observed a 79% specificity when evaulating the model on EEG-window level. When aggregating predictions to recording level, the specificity drops to 48%. These results demonstrate the potential of transformer-based models to provide scalable EEG screening in resource-limited settings.



References

[1] W.-B. Jiang and others, "Large Brain Model for Learning Generic Representations with Tremendous EEG Data in BCI," 2024.

[2] A. Gjølbye, W. Lehn-Schiøler, and others, "Speed: Scalable Preprocessing of EEG Data for Self-Supervised Learning," 2024.

[3] S. Beniczky and others, "Standardized computer-based organized reporting of EEG: SCORE – Second version," 2017.